



운영체제의 기본 옵션에는 컨테이너가 있어요

UST - ETRI 석사과정 인공지능학과
차주형

발표 순서 및 개요

1. 공유기와 스마트폰에서 딥 러닝 모델 연산을 수행해봤다! (1분)
2. 원리 설명 및 분석 (3분)
3. 이것 이용하여 무엇을 더 할 수 있을까? (1분)

EXPERIMENTAL SETUP

실험 환경

IPTIME A1004NS



ASUS RT-AX53U



Q. SD 865 HDK (Galaxy S20)



OS : BusyBox v1.8.2

CPU : MT7620A

Arch : MIPSEL (RISC-V)

Experiments : QEMU

OS : OpenWRT 23.05.5

CPU : MT7621AT

Arch : MIPSEL (RISC-V)

Experiments : SSH

OS : Android 10

CPU : Q. SD 865

Arch : ARM

Experiments : ADB

IPTime A1004NS

1. FLASH ROM의 용량이 16MB, 운영체제 용량이 12.6MB로 추가적인 소프트웨어 설치 자체가 불가능 하였음
2. 가상 머신 기반으로 IPTime A1004NS 펌웨어 1.2.152 버전으로 실험을 수행함
3. 펌웨어에 chroot가 기본 내장되어 있었기 때문에 데비안 컨테이너를 생성이 가능하였음
4. 외장 디스크를 통해 공간의 확충이 가능하다면 실제 하드웨어에서 컨테이너 생성의 가능성을 시사함

데비안 컨테이너 생성 결과

```
(base) chacha@chacha:~/iptime$ sudo chroot . ./qemu-mipsel-static ./bin/ash
BusyBox v1.8.2 (2022-05-30 15:03:58 KST) built-in shell (ash)
Enter 'help' for a list of built-in commands.

# chroot ~/mips-rootfs
root@chacha:~# apt
apt 1.0.9.8.4 for mips compiled on Dec 11 2016 09:52:19
Usage: apt [options] command

CLI for apt.
Basic commands:
list - list packages based on package names
search - search in package descriptions
show - show package details
```

컨테이너와 Python에서 LeNet 연산

```
root@chacha:~# python3
Python 3.11.0 (main, Nov 11 2024, 16:32:55)
Type "help", "copyright", "credits" or "license()" for more
>>> exit()
root@chacha:~# python3 ./a.py
0.012451410293579102
0.011252641677856445
0.011307954788208008
0.011332988739013672
0.01131892204284668
0.011331796646118164
```

ASUS RT-AX53U

1. 저장 공간의 부족하였지만, USB 를 통해 실제 하드웨어에서 실험을 진행함
2. 펌웨어에 chroot가 기본 내장되어 있었기 때문에 데비안 컨테이너를 생성이 가능하였음
3. MIPSEL(RISC-V)에서 Python 환경에서 LeNet 모델 연산과 패키지 매니저가 동작함을 확인하였음.
4. LeNet 모델에 대해 1회 추론에 37.3ms 로 얻을 수 있었음.

데비안 컨테이너 생성 결과

```
(base) chacha@chacha:/$ ssh etri@129.254.196.83 -p 3333
etri@129.254.196.83's password:
etri@RT-AX53U-3600:/tmp/home/root# cd /mnt/sdb1/
etri@RT-AX53U-3600:/tmp/mnt/sdb1# chroot ./mips-rootfs
root@RT-AX53U-3600:/# apt

apt 1.0.9.8.4 for mips compiled on Dec 11 2016 09:52:19
Usage: apt [options] command

CLI for apt.
Basic commands:
list - list packages based on package names
search - search in package descriptions
show - show package details

update - update list of available packages
```

컨테이너와 Python에서 MNIST 연산

```
etri@RT-AX53U-3600:/tmp/mnt/sdb1# python3
Python 3.11.0 (main, Nov 11 2024, 16:32:55) [GCC 10.3.0] on linux
Type "help", "copyright", "credits" or "license" for more information.
>>> exit()
etri@RT-AX53U-3600:/tmp/mnt/sdb1# python3 ./lenet.py
0.4360527992248535
0.37165212631225586
0.37251734733581543
0.37381982803344727
0.3729710578918457
0.37374138832092285
0.3733081817626953
0.3731210231781006
0.3738255500793457
```

Qualcomm SD 865 HDK

1. 데비안 계열 컨테이너를 생성하여 NumPy와 PyTorch를 통한 딥 러닝 연산이 가능하였음
2. **자바 없이**, OpenCL / GPU 연산, 리눅스 GUI 개발이 **파이썬과 C++**로 가능하였음
3. 컨테이너 상에서 하드웨어 제어와 가속이 가능하였으며, 하나의 시스템으로 확장이 가능하였음

데비안 컨테이너 생성 결과

```
> adb shell
kona:/ # cd /data/local/tmp
kona:/data/local/tmp # ls
demo start-ubuntu20.sh ubuntu20-binds ubuntu20-fs
kona:/data/local/tmp # sh ./start-ubuntu20.sh
root@localhost:~# apt
apt 2.0.9 (arm64)
Usage: apt [options] command

apt is a commandline package manager and provides commands for
searching and managing as well as querying information about packages.
It provides the same functionality as the specialized APT tools,
like apt-get and apt-cache, but enables options more suitable for
interactive use by default.

Most used commands:
  list - list packages based on package names
  search - search in package descriptions
  show - show package details
  install - install packages
```

컨테이너와 Python에서 MNIST 연산

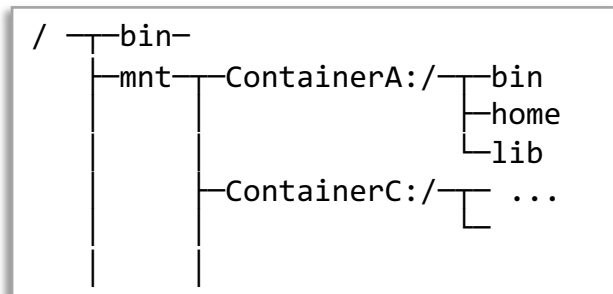
```
Python 3.8.10 (default, May 26 2023, 14:05:08)
[GCC 9.4.0] on linux
Type "help", "copyright", "credits" or "license" for more information.
>>> exit()
root@localhost:~# python3 ./lenet.py
10.58 ms
4.14 ms
2.88 ms
2.93 ms
2.87 ms
2.82 ms
2.91 ms
2.75 ms
2.17 ms
2.27 ms
```

컨테이너의 주요 핵심 기능인 Chroot가 도대체 뭐야?

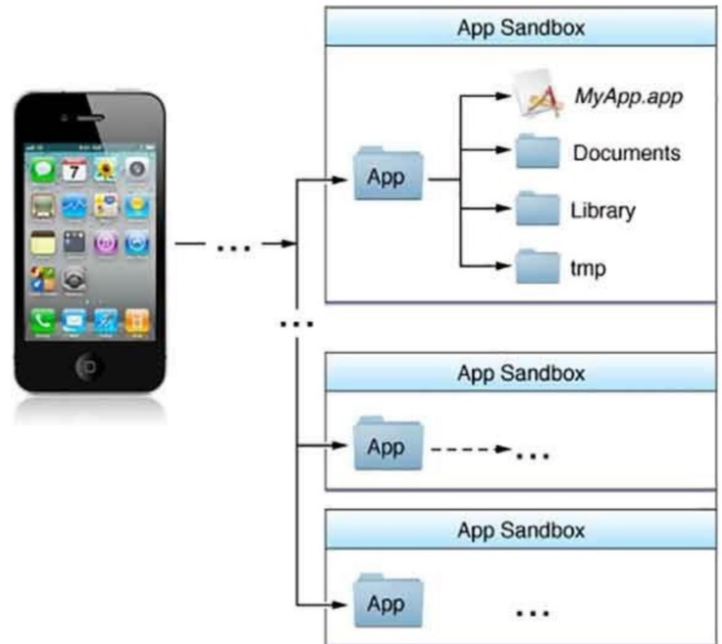
1. Chroot는 Docker, Podman에서 사용되며 가상환경, 샌드박스 기능을 제공함.
2. 그리고, 내부 구현은 단순히 **“루트 디렉토리”**의 경로만 옮기는 방식으로 구현됨.
3. 루트 디렉토리에서 벗어나는 행위를 “탈옥” 이라고 이야기하며, ! “아이폰 탈옥” 어원이 이곳 !

컨테이너

Chroot (저장소의 격리)



=



왜 이것이 가능할까?

1. BusyBox, Android, OpenWRT 의 빌드 할 때 기본 옵션값이 Chroot는 활성화되어 있음.
2. 제조 회사가 따로 빌드 옵션을 바꾸지 않는 이상 기본적으로 Chroot가 존재함.
3. 우리 주변에 널려있는 공유기는 최고의 리눅스 머신입니다.

```
BusyBox 1.37.0 Configuration

Convertile
Arrow keys navigate the menu. <Enter> selects submenus ----. Highlighted letters are hotkeys. Pressing <Y> includes, <A> excludes,
<M> modularizes features. Press <Esc><Esc> to exit, <?> for Help, </> for Search. Legend: [*] built-in [ ] excluded <A> module
< > module capable

[*] Support verbose options (usually -v) for various applets (NEW)
--- Common options for date and touch
[*] Allow timezone in dates (NEW)
--- Common options for cp and mv
[*] Preserve hard links (NEW)
--- Common options for df, du, ls
[*] Support human readable output (example 13k) (NEW)
[*] basename (3.7 kb) (NEW)
[*] cat (5.8 kb) (NEW)
[*] Enable -n and -b options (NEW)
[*] cat -v[etA] (NEW)
[*] chgrp (7.6 kb) (NEW)
[*] chmod (5.5 kb) (NEW)
[*] chown (7.6 kb) (NEW)
[*] Enable long options (NEW)
[*] chroot (4 kb) (NEW)
[*] cksum (4.3 kb) (NEW)
[*] crc32 (4.2 kb) (NEW)
[*] comm (4.4 kb) (NEW)
[*] cp (10 kb) (NEW)
[*] Enable long options (NEW)
[*] Enable --reflink[=auto] (NEW)
[*] cut (6.7 kb) (NEW)
```

BusyBox 빌드 환경

OpenWRT 빌드 환경

```
.config - OpenWrt Configuration
> Base system > Coreutils

Coreutils
Arrow keys navigate the menu. <Enter> selects submenus ---- (or empty submenus ----). Highlighted letters are
<Y> includes, <A> excludes, <M> modularizes features. Press <Esc><Esc> to exit, <?> for Help, </> for Search.
[ ] excluded <A> module < > module capable

[ ] Support verbose options (usually -v) for various applets (NEW)
*** Common options for date and touch ***
*** Common options for cp and mv ***
[*] Preserve hard links (NEW)
: *** Common options for df, du, ls ***
[*] Support human readable output (example 13k, 23M, 235G) (NEW)
[*] basename (438 bytes) (NEW)
[*] cat (5.6 kb) (NEW)
[ ] Enable -n and -b options (NEW)
[ ] cat -v[etA] (NEW)
[*] chgrp (7.6 kb) (NEW)
[*] chmod (5.5 kb) (NEW)
[*] chown (7.6 kb) (NEW)
[ ] Enable long options (NEW)
[*] chroot (3.7 kb) (NEW)
[ ] cksum (4.1 kb) (NEW)
[ ] crc32 (4.1 kb) (NEW)
[ ] comm (4.2 kb) (NEW)
[*] cp (10 kb) (NEW)
[ ] Enable long options (NEW)
[*] cut (5.8 kb) (NEW)
[ ] cut -F (NEW)
[*] date (7 kb) (NEW)
```


HOW?

Docker Hub에 있는 모든 이미지를 가져와서 실험 할 수 있습니다.

! 컨테이너 이미지 저장소 !



다운로드



컨테이너 실행 !!!



HOW?

Docker Hub에 있는 모든 이미지를 가져와서 실험 할 수 있습니다.

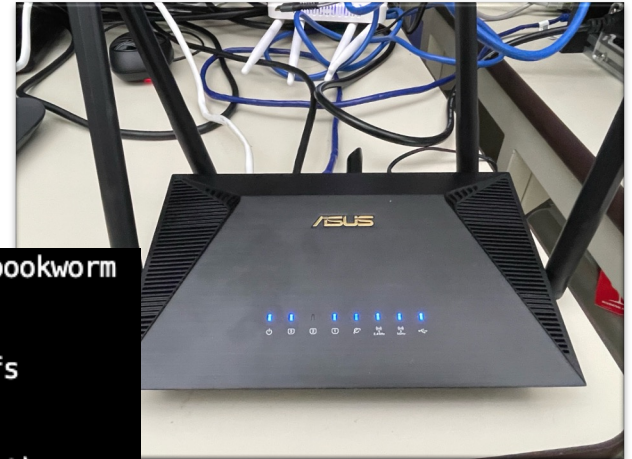
! 컨테이너 이미지 저장소 !



다운로드



컨테이너 실행 !!!



```
chacha@compute:~/test$ docker create --name tmpimg golang:1.25.1-bookworm
03e64fdcf834de8266f76f3d41c34de36d23ca00ae01e59b436668006ec971e7
chacha@compute:~/test$ docker export tmpimg > rootfs.tar
chacha@compute:~/test$ mkdir rootfs && tar -xf rootfs.tar -C rootfs
chacha@compute:~/test$ sudo chroot ./rootfs/
[sudo] password for chacha:
bash: warning: setlocale: LC_ALL: cannot change locale (en_US.UTF-8)
root@compute:/# PATH=/go/bin:/usr/local/go/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin
root@compute:/# go
failed to start telemetry sidecar: os.Executable: readlink /proc/self/exe:
Go is a tool for managing Go source code.
```

Usage:

go <command> [arguments]

The commands are:

bug	start a bug report
build	compile packages and dependencies
clean	remove object files and cached files
doc	show documentation for package or symbol

HOW?

Docker Hub에 있는 모든 이미지를 가져와서 실험 할 수 있습니다.

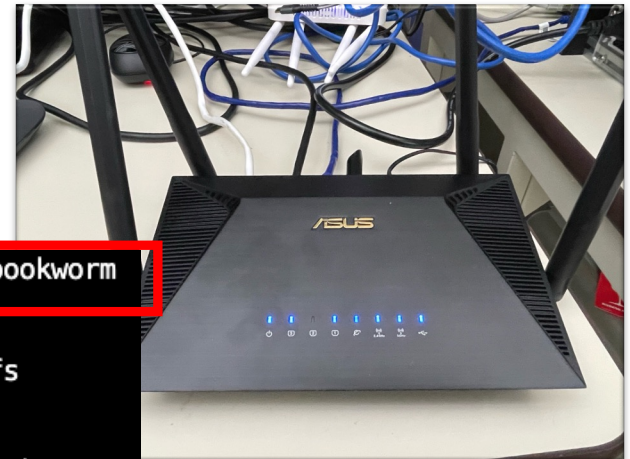
! 컨테이너 이미지 저장소 !



다운로드



컨테이너 실행 !!!



```
chacha@compute:~/test$ docker create --name tmpimg golang:1.25.1-bookworm
03e64fd8c834de8266f76f3d41c34de36d23ca00ae01e59b436668006ec971e7
chacha@compute:~/test$ docker export tmpimg > rootfs.tar
chacha@compute:~/test$ mkdir rootfs && tar -xf rootfs.tar -C rootfs
chacha@compute:~/test$ sudo chroot ./rootfs/
[sudo] password for chacha:
bash: warning: setlocale: LC_ALL: cannot change locale (en_US.UTF-8)
root@compute:/# PATH=/go/bin:/usr/local/go/bin:/usr/local/sbin:/usr/local/bin
root@compute:/# go
failed to start telemetry sidecar: os.Executable: readlink /proc/self/exe:
Go is a tool for managing Go source code.

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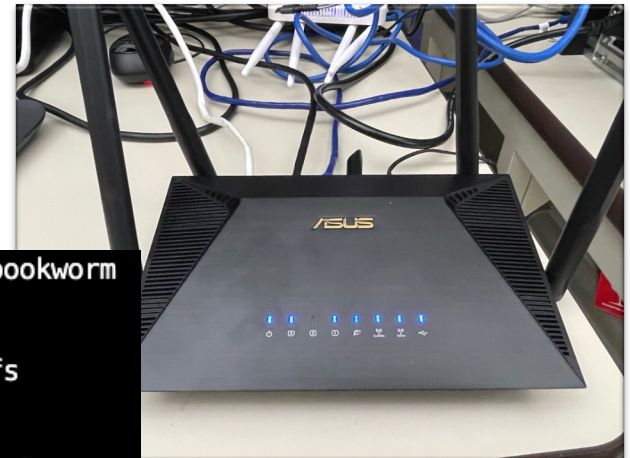
! 컨테이너 이미지 저장소 !



다운로드



컨테이너 실행 !!!



```
chacha@compute:~/test$ docker create --name tmping golang:1.25.1-bookworm
03e64fd8c834de826617615d41c34de30d25ca00ae01e39b430000000e971e7
chacha@compute:~/test$ docker export tmping > rootfs.tar
chacha@compute:~/test$ mkdir rootfs ff tar -xvf rootfs tar -C rootfs
chacha@compute:~/test$ sudo chroot ./rootfs/
[sudo] password for chacha:
bash: warning: setlocale: LC_ALL: cannot change locale (en_US.UTF-8)
root@compute:/# PATH=/go/bin:/usr/local/go/bin:/usr/local/sbin:/usr/local/b
root@compute:/# go
failed to start telemetry sidecar: os.Executable: readlink /proc/self/exe:
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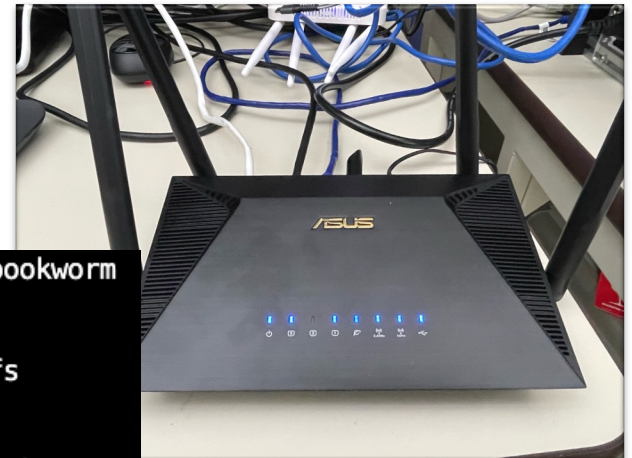
! 컨테이너 이미지 저장소 !



다운로드



컨테이너 실행 !!!



```
chacha@compute:~/test$ docker create --name tmpimg golang:1.25.1-bookworm
03e64fd8c834de8266f76f3d41c34de36d23ca00ae01e59b436668006ec971e7
chacha@compute:~/test$ docker export tmpimg > rootfs.tar
chacha@compute:~/test$ mkdir rootfs && tar -xvf rootfs.tar -C rootfs
chacha@compute:~/test$ sudo chroot ./rootfs/
[sudo] password for chacha.
bash: warning: setlocale: LC_ALL: cannot change locale (en_US.UTF-8)
root@compute:/# PATH=/go/bin:/usr/local/go/bin:/usr/local/sbin:/usr/local/bin
root@compute:/# go
failed to start telemetry sidecar: os.Executable: readlink /proc/self/exe:
Go is a tool for managing Go source code.
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doc	show documentation for package or symbol

HOW?

Docker Hub에 있는 모든 이미지를 가져와서 실험 할 수 있습니다.

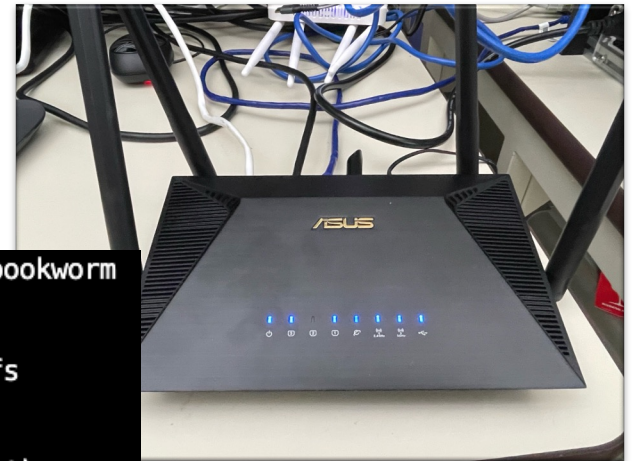
! 컨테이너 이미지 저장소 !



다운로드



컨테이너 실행 !!!



```
chacha@compute:~/test$ docker create --name tmpimg golang:1.25.1-bookworm
03e64fd8c834de8266f76f3d41c34de36d23ca00ae01e59b436668006ec971e7
chacha@compute:~/test$ docker export tmpimg > rootfs.tar
chacha@compute:~/test$ mkdir rootfs && tar -xf rootfs.tar -C rootfs
chacha@compute:~/test$ sudo chroot ./rootfs/
[sudo] password for chacha:
bash: warning: setlocale: LC_ALL: cannot change locale (en_US.UTF-8)
```

```
root@compute:/# PATH=/go/bin:/usr/local/go/bin:/usr/local/sbin:/usr/local/
root@compute:/# go
failed to start telemetry sidecar: os.Executable: readlink /proc/self/exe:
Go is a tool for managing Go source code.
```

Usage:

go <command> [arguments]

The commands are:

bug	start a bug report
build	compile packages and dependencies
clean	remove object files and cached files

LinkedIn



CV



THANK YOU

UST 석사과정 인공지능학과

발표자 : 차주형

chacha@udon.party



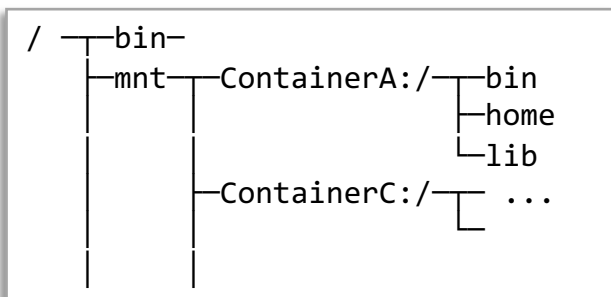
Container는 3개의 핵심 요소로 이뤄져 있음.

1. 컨테이너 표준안(OCI) 에는 컨테이너와 정보 보안 및 격리 기능이 포함됨.
2. 컨테이너 기능만 이용하고 싶다면, Chroot, 저장소의 격리만 활용하면 됨.
3. 즉, 공유기에 우분투를 올리고 싶다면 Chroot만 사용하면? **완벽**



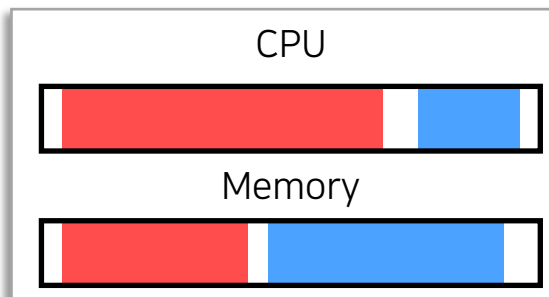
컨테이너

Chroot (저장소의 격리)

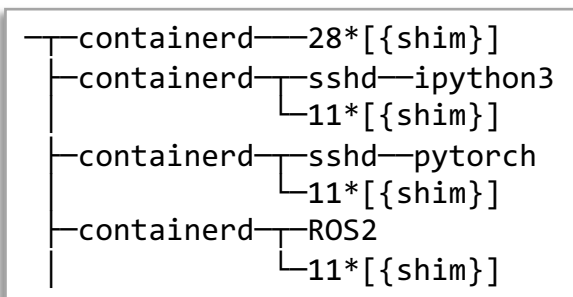


정보 보안 및 격리

CGroups (하드웨어 자원 격리)



Namespace (프로세스간 통신 격리)



References and Sources

[1] Sandbox Images : <https://www.simplykyra.com/blog/how-to-debug-and-solve-an-app-sandbox-related-issue-within-the-xcode-ide/>